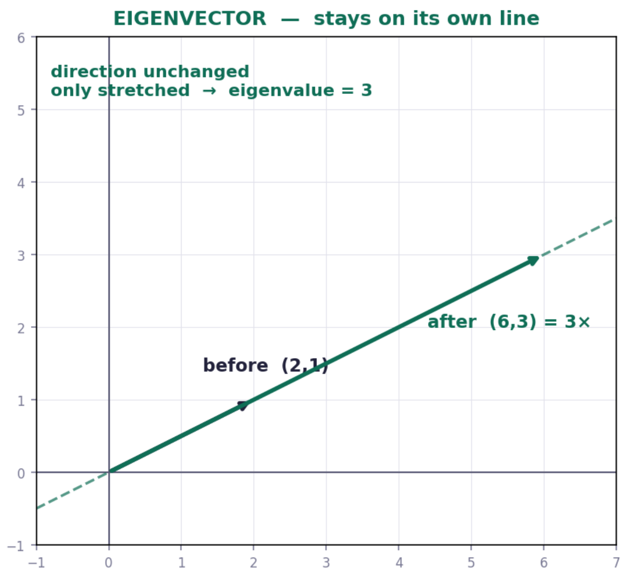
Eigenvectors and eigenvalues

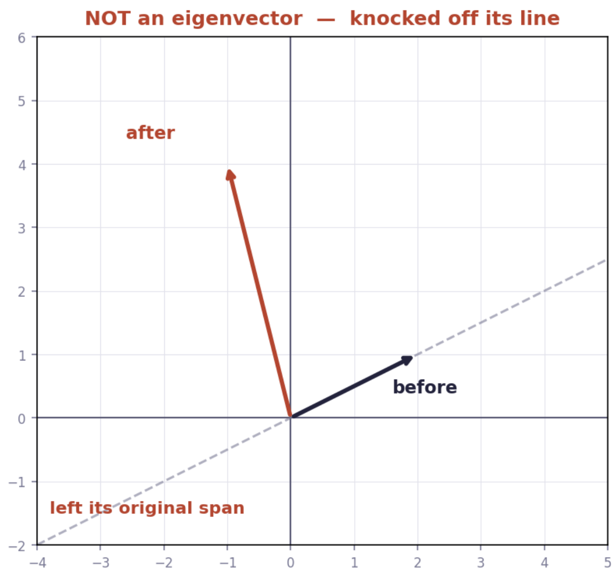
The special directions a transformation cannot knock off course.

THE IDEA

Most vectors get rotated off their original line by a transformation. But some special vectors stay on their own span — they only get **stretched or squashed**. Those are **eigenvectors**, and the stretch factor is the **eigenvalue** λ.

> **The defining equation:** **A v = λ v** — applying the matrix does the same thing as multiplying by a single number.





HOW TO FIND THEM

Step	What to do
1	Write A − λI (subtract λ from each diagonal entry)
2	Set **det(A − λI) = 0** and solve for λ — these are the eigenvalues
3	For each λ, solve (A − λI)v = 0 to get the eigenvector direction
4	Check: A v should equal λ v

Why det = 0? Because A − λI must squash v to zero, and only a collapsing transformation (Chapter 6) can send a non-zero vector to zero.

WORKED EXAMPLE

CHAPTER 14 — WORKED EXAMPLE

find the eigenvalues and eigenvectors of a 2×2 matrix

THE EQUATION

$$A\,v = \lambda\,v \qquad \det(A - \lambda I) = 0$$

λ (lambda) = the eigenvalue = the stretch factor

STEP 1 — write $A - \lambda I$

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \qquad A - \lambda I = \begin{bmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix}$$

STEP 2 — set the determinant to zero

$$\det = (3-\lambda)(2-\lambda) - (1)(0) = 0$$

$$(3-\lambda)(2-\lambda) = 0$$

$$\lambda = 3 \quad \text{or} \quad \lambda = 2$$

these are the eigenvalues

STEP 3 — find the eigenvector for $\lambda = 3$

solve $(A - 3I)\,v = 0$:

$$\begin{matrix} 0 & 1 \\ 0 & -1 \end{matrix} \times (x, y) = (0, 0)$$

row 1 : $0x + 1y = 0 \rightarrow y = 0$

x is free $\rightarrow$ take $x = 1$

eigenvector = (1, 0)

STEP 4 — check it

$$A\,(1, 0) = 1\times(3, 0) + 0\times(1, 2) = (3, 0) = 3 \times (1, 0) \quad ✓$$

ML link : eigenvalues drive PCA, stability of training, and how gradients grow or vanish in deep networks.

Why it matters for ML: eigenvalues are behind PCA (the principal components are eigenvectors of the covariance matrix), the stability of training, and why gradients explode or vanish in deep networks. This is the most ML-relevant chapter in the series.

TEST — Chapter 14

1. A has columns (2, 0) and (0, 5). Find both eigenvalues. → **$\lambda = 2$ and $\lambda = 5$ (diagonal matrix)**
2. A has columns (3, 0) and (1, 2). Find both eigenvalues. → **$(3-\lambda)(2-\lambda) = 0 \rightarrow \lambda = 3, \lambda = 2$**

3. Verify that (1, 0) is an eigenvector of the matrix with columns (4, 0) and (7, 3), and give its eigenvalue. **→ A(1,0) = (4,0) = 4(1,0) → λ = 4**

FORMULAS FROM THIS CHAPTER

Definition	A v = λ v
Find λ	det(A − λI) = 0
Find v	solve (A − λI)v = 0
Check	A v must equal λ v